MILITARY SPECIFICATION

INDICATOR, HORIZONTAL SITUATION, AQU-2/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers the requirements for one type of horizontal situation indicator, designated AQU-2/A.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified within.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation-packaging, Methods Of
MIL-C-675	Coating Of Glass Optical Elements (Antireflection)
DOD-D-1000	Drawings, Engineering And Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For

Add the following boxed paragraph to the bottom center of page:

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

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MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-L-25467	Lighting, Integral, Red, Aircraft, Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
MIL-C-83488	Coatings, Aluminum, ION Vapor Deposited

STANDARDS

Federal

FED-STD-595	Color

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-704	Aircraft Electrical Power Characteristics
MIL-STD-826	Electromagnetic Interference, Test Requirements And Test Methods
MIL-STD-838	Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4).

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3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable porperties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exceptions in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt, spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.6 Corrosive fumes. The materials as installed in the HSI and under the service conditions specified herein shall not liberate deleterious fumes.

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3.3.7 Hermetic seal. Nonferrous materials contained within hermetically sealed inclosures (see 6.4.2), shall be considered suitably protected from corrosion. Requirements specified for fungicidal and corrosion protective treatment and anodizing of aluminum alloy parts shall not be applicable for parts within hermetically sealed inclosures. Steel parts within such inclosures shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.3.8 Protective treatment. When materials are used in the construction of the HSI that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.9 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

3.4 Design and construction.

3.4.1 Design. The HSI shall be designed to present the following information:

a. Stabilized magnetic compass heading provided from the master directional reference

b. Command heading information provided from a remote source or desired heading, set in manually

c. Command course provided from a remote source or desired course, set in manually, displayed by a pointer and a numerical indication

d. Course bar which indicates displacement of the aircraft from a manually or automatically selected track

e. TO-FROM indication from a selected radio facility

f. Numerical indication of distance to a transmitting radio

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source or target; indication masked when not in use

g. Bearing to selected station or target

3.4.2 Construction. The HSI shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The HSI shall be so constructed that overhaul, including replacement of worn parts, adjustment balance, and calibration may be accomplished by instrument and electronic repairmen provided with simplified tools and equipment, and trained in the repair of electronic equipment.

3.5 Performance. The HSI shall conform to all requirements specified herein after exposure to the following conditions:

a. Operating temperature ranging from -54° to +71°C and storage temperatures ranging from -65° to +85°C

b. Altitudes from sea level to 100,000 feet

c. Relative humidity up to 100 percent

d. Fungus growth as encountered in tropical climates

e. Exposure to salt-laden atmosphere

f. Sand and dust particles as encountered in desert areas

g. Thermal shock forces as encountered during rapid changes of extreme temperature

h. Vibration forces incident to service use

i. Immersion in water and leak test conducted with an approved helium leak detector, without evidence of leakage in excess of that specified herein

j. Acceleration forces as encountered in high-performance aircraft

k. The application of 250V DC for 10 seconds

3.5.1 Radio interference. The HSI shall not be a source of radio interference and shall meet the requirements of MIL-STD-826.

3.5.2 Attitude. The HSI shall be capable of operating

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satisfactorily while in any attitude.

3.5.3 Friction error. A minimum of error shall be introduced in the readings due to the friction between moving parts.

3.5.4 Magnetic effect. The HSI shall not cause excessive error in the indication of panel-mounted directional magnetic compasses.

3.5.5 Life. The HSI shall meet the performance requirements for 1,000 hours operation during actual service usage.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

3.7 Electronic requirements. The requirements specified in MIL-E-5400 are applicable as electronic requirements of this specification. Additional requirements shall be as specified herein.

3.7.1 Requirements conflict. In the event the requirements of MIL-E-5400 and this specification conflict, the requirements of this specification shall govern.

3.8 Miniaturization. Modern techniques of miniaturization, potting, and hermetic sealing shall be exploited to the greatest extent practicable without sacrificing ruggedness, reliability, and service life.

3.8.1 Transistorized or magnetic amplifier circuits. Four transistorized or magnetic servo amplifiers shall be integral with the HSI mechanism within the HSI case. The amplifiers shall feature module plug-in capability, if possible. Each amplifier module shall be designed with partial redundant circuits so that the servos will continue to operate, though with reduced performance, should partial failure of the amplifier occur.

3.9 Case. The HSI case shall be in accordance with figure 1. The dimensions of the cover glass aperture shown on figure 2 shall apply.

3.9.1 Control knobs. Control knobs as shown on Figure 1 shall be provided. The knobs shall be medium gray, color No. 36231 of FED-STD-595. The lettering of the knobs shall be block type. The inside of the case around the electrical connector shall be of sufficient strength to prevent damage to the case when the

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electrical connector is tightened during installation. The case shall be metal, uniform in texture with a smooth surface, and entirely covered with a durable finish. The ends of the knobs shall be black with white lettering. A minimum of 2 inch ounces and a maximum of 10 inch ounces shall be required to turn the knobs. Sealed fluid couplings may be utilized for the control knobs.

3.9.2 Cover glass. The cover glass shall be hermetically sealed within the HSI case and shall be mounted from the inside of the case. The cover glass shall be from 3/16 to 1/4 inch in thickness. It shall be flat and free from discolorations, except as induced by glare reducing coating, scratches, and striae which interfere with reading of the instrument. The cover glass aperture shall be as specified on figure 1.

3.9.3 Anti-reflection coating. A magnesium fluoride coating in accordance with MIL-C-675 shall be applied to each surface of the cover glass and the lighting wedge. Coated surfaces shall not chip, scale, peel, flake, dissolve, discolor, or otherwise be adversely affected in normal service use. The coating on the outer surface of the cover glass shall withstand abrasion incident to normal service usage.

3.9.4 Hermetic sealing. The case shall be hermetically sealed. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the component may be subjected. The case shall be designed so that is will meet the necessary pressure requirements when tested for leaks by means of a mass-spectometer-type of helium leak detector.

3.9.4.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. Where practicable, the 100 percent helium filling medium shall be utilized.

3.9.4.2 Leakage rate. The indicator shall be sealed so that the initial maximum leakage rate at a pressure differential of approximately 1 atmosphere will not permit more than 100 percent

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loss of the total filling medium after 1,000 hours.

3.10 HSI display. The display detail dimensions shall conform to one of the following design criteria options:

3.10.1 Option A. The HSI display shall conform to figures 2 and 4. All sizes of the pointers, azimuth card, deviation bar, numerals, and letters shall conform to the size and width listed on figures 2 and 4. The azimuth card, center section of the display, and main mask of the display shall lie in one plane, designated plane A, and shall be 0.7 ±0.05 inch from the forward edge of the cover glass. The forward edge of the heading marker shall be 0.01 inch forward of plane A; the forward face of the bearing pointer shall be 0.03 inch forward of plane A; the lubber line shall be 0.05 inch forward of plane A; the forward face of the course arrow and deviation bar shall be 0.18 inch forward of plane A; and the fixed airplane reference shall be 0.25 inch forward of plane A. The first set of deviation dots shall be 0.11 inch forward of plane A and the second set shall be 0.03 inch forward of plane A. The forward face of the TO-FROM arrow shall be not more than 0.025 inch behind plane A. The dimensions as given are design center values.

3.10.2 Option B. The HSI display shall conform to figure 2. All sizes of the pointers, azimuth card, deviation bar, numerals, and letters shall conform to the size and width indicated on figure 2 and in table I. The azimuth card, main mask of the display, course arrow, and deviation bar, shall all lie on the same plane designated plane A and shall be 0.5 ±0.05 inch from the forward edge of the cover glass. The airplane and lubber line shall be 0.05 inch forward of plane A. The forward face of the bearing pointer shall be 0.03 inch forward of plane A. The forward face of the heading marker shall be 0.01 inch forward of plane A. The forward face of the TO-FROM arrow shall be 0.12 inch back of plane A. The second set of deviation dots shall be 0.13 inch back of plane A. The deepest point of any visible portion of the display shall be 0.17 inch from plane A. These dimensions as given are design center values.

3.11 Standard synchro test transmitter. A standard calibrated synchro with minimum and known errors shall be provided and shall be an Eclipse-Pioneer Type AY201-1, or equal, high precision transmitting synchro. A 0° to 360° dial, clamped to the rotor of the synchro, shall be settable, rotatable, and readable through 360° of rotation to 6 minutes of arc. The procedure in 3.11.1 shall be used to calibrate the test synchro.

3.11.1 Calibrating the test sunchro. The rotor leads shall be designated as H and C and one stator lead designated as Z and

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connected to C. When 26V is applied to H and C (C being ground) the dial shall be positioned on 0° and clamped to the rotor when:

a. The voltage across H and Y is maximum.

b. The voltage across the remaining two stator leads (X and Y) is minimum.

c. The voltage across X and Z increases before it decreases for increasing heading indication of the dial. The voltage X to Z and Y to Z shall be in phase with the excitation voltage C to H when the dial reading is zero. For increasing heading indication of the dial, the voltage X to Y shall increase and be in phase with the excitation voltage C to H. The test transmitter will be set at an index reference of zero and a positive rotation reference XYZ.

3.12 Presentation. The presentation shall be in accordance with figure 2 and shall consist of a rotating-card type azimuth ring, a lubber line, a heading marker, a course arrow, a TO-FROM arrow, a course bar, a bearing pointer, a SELECTED COURSE window, a distance window, and a miniature airplane symbol behind the cover glass of the instrument to present a pictorial display of the navigation situation.

3.13 Lubber line and airplane. The lubber line and airplane symbol shall be in accordance with figure 2.

3.14 Aircraft heading. The aircraft heading shall be displayed on the azimuth ring and shall be read under the fixed lubber line located at the top of the indicator face as shown on figure 2. The azimuth ring shall be in accordance with figure 2.

3.14.1 Azimuth ring signal input. The input synchro CT (control transformer) shall be a high impedance type EP AY500-5, or equal, which shall receive an input from a synchro device similar to the test transmitter described in 3.11. A size 8 synchro CT, with the general impedance of the stator windings as a type EP-AY500-5, may be utilized.

3.14.2 Azimuth ring servo system. A servo follow-up system consisting of a CT, motor, rate generator feedback, and amplifier shall drive the azimuth card and rotor or the input CT to a null. It shall be capable of receiving a change of heading at any rate between 0° and 60° per second without hunting. There shall be no appreciable hunt or oscillation in the servo system while the input compass signal is changing or stationary. The servo system shall position the heading indication under lubber line to within ±1° of the position of a calibrated synchro. The

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design of the azimuth card gearing shall result in operation such as would be achieved in the system shown on figure 3. The electrical pin connections of the azimuth card servo system shall conform to that shown on figure 3.

3.14.3 Azimuth card control transformer zero. The azimuth card CT shall be zeroed with a positive rotation reference at an index reference of 270°. With the XYZ leads of the azimuth card CT as specified on figure 3 and the excitation voltage of the test transmitter C to H in phase with the power input pin AA to pin BB:

a. The azimuth ring shall position at 0° (north) when the rotor of the test transmitter dial is at 0°.

b. If the dial on the rotor of the test transmitter is continuously rotated so that its heading indication increases, the azimuth ring heading shall increase (ccw rotation) accordingly and the voltage C to H of the CT shall be in phase with the excitation voltage C to H.

3.15 Command heading. A means of selecting a desired heading both automatically and manually shall be provided. A heading marker as shown on figure 2, which can rotate around the azimuth ring, shall be driven by a servo system when excited by a remote device similar to the standard test transmitter specified in 3.11 and manually by a knob at the lower left corner of the instrument. There shall be no visible interaction between the knob and the servo motor so that when the remote synchro device is driving the servo system, turning of the knob will have no lasting effect on the heading marker position. The heading marker shall rotate in synchronism with the azimuth ring after it is set. The heading selection knob shall be connected with the heading marker in such a manner that cw (clockwise) rotation of the knob produces cw rotation of the heading marker, and vice versa. The turning ratio between the knob and the heading marker shall be between 8 and 10 to 1. The azimuth scale reading indicated under the heading marker shall be the desired heading. The heading set knob shall be as shown on figure 1. The maximum backlash between the command heading marker and the azimuth card shall be 1/4°.

3.15.1 Heading marker servo system. A servo system similar to that described in 3.14.2 for driving the azimuth ring shall be employed for positioning the heading marker in response to a remote input signal. The servo shall receive a change of command heading at any rate between 0° and 60° per second without hunting. There shall be no appreciable hunt or oscillation in the servo system and it shall position the heading marker around

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the azimuth ring to within ±1° of the position of the test synchro. The design of the differential gearing shall result in operation of the heading marker with respect to the azimuth ring, remote synchro command input, and knob command input as that achieved on figure 3. The heading marker servo shall conform to the electrical connector and pin wiring shown on figure 3.

3.15.2 Heading marker signal input. The heading marker servo system shall receive its input signal from a transmitting synchro device similar to the test transmitter described in 3.11. The heading synchro device shall be a Clifton Precision Transolver CDSH-10-AS-4, or equivalent, capable of operating as a CT and a transmitter as shown on figure 3. The XYZ load shall be equivalent to an EP AY500-3 or one exhibiting higher impedance characteristics.

3.15.3 Heading datum control transformer. The heading datum CT shall be Clifton Precision's CTC-8-A-4, or equivalent, and shall operate as shown on figure 3.

3.15.4 Heading command and heading datum control transformer zero. Both the heading command and heading datum CT's shall be zeroed with positive rotation referenced at an index reference of 270°. Both synchros shall be zeroed in the following manner.

a. With the standard test transmitter positioned to zero indication on its dial and locked, and connected to the heading command and heading datum CT's as per the XYZ notation indicated on figure 3, the standard test transmitter shall be excited with 26V 400 cycle.

b. The heading marker shall be positioned to 0° (north) on the azimuth card and the voltage C to H' of the heading command CT rotor and heading datum CT rotor as shown on figure 3 shall be null.

c. If the heading marker is rotated cw around the azimuth ring, the voltage C to H' of the heading command CT rotor and heading datum CT rotor shall be in the opposite phase sense as the voltage C to H of the standard test transmitter from 0° (north) to 180°, null at 180° (south) and shall be in the same phase sense as the voltage C to H of standard test transmitter from 180° (south) to 0° (north).

d. If the heading marker is set on 0° (north) on the azimuth dial and if the rotor of the standard test transmitter is rotated to indicate an increasing heading (dial ccw), the voltage C to H' of the heading command CT and the heading datum CT shall be null at 0°, in the same phase sense as the excitation voltage

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C to H from 0° to 180°; null at 180°, and opposite to the phase sense of C to H from 180° to 0°.

e. If the transmitter winding of the heading command CT is excited with 26V ac so that the voltage C to H is in phase with power excitation ground to phase and the heading marker is set on 0° (north), then the voltage X to Y and Y to Z will be in phase with C to H and voltage X to Y null. If the heading marker is rotated cw to (east), the voltage X to Z will increase before decreasing.

3.15.5 Heading marker servo zero. The heading command control transformer shall be connected to its servo system. The standard test transmitter shall be connected to the heading command CT as per the XYZ notation on figure 3. The excitation voltage of the test transmitter C to H shall be in phase with the servo power excitation voltage, pin AA to pin CC. The following conditions shall be satisfied:

a. The heading marker shall position to 0° (north) on the azimuth ring when the dial of the standard test transmitter is at 0°.

b. The heading marker shall rotate in an increasing heading indication on the azimuth ring (cw) when the dial of the test transmitter synchro is rotated in an increasing heading indication (dial ccw).

c. If the dial of the standard test transmitter is rotated in the increasing heading indication, the voltage C to H' of the heading command CT shall be in the same phase sense with the voltage C to H of the standard test transmitter.

3.16 Command course. A means of selecting a desired course both automatically and manually shall be provided. A course arrow, as shown on figure 2, that can rotate around the inside edge of the azimuth ring shall be driven by a servo system when excited by a remote device similar to the standard test transmitter specified in 3.11 and shall be manually turned by a knob at the lower right corner of the instrument. As in the heading marker servo system, turning of the course knob shall have no lasting effect on the course arrow when the servo is being driven by a remote device. The course marker shall rotate in synchronism with the azimuth ring after it is set. The scale reading on the azimuth ring at the tip of the course arrow shall be the selected course or track. The course selection knob shall be connected with the course arrow in such a manner that cw rotation of the knob will produce cw rotation of the course arrow, and vice versa. The turning ratio between the knob and the course marker shall be

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between 8 to 10 to 1. The course set knob shall be as shown on figure 2. A digital-type counter shall present the selected course in a window on the right side of the HSI.

3.16.1 Course marker servo system. A servo system similar to that described in 3.14.2 for driving the azimuth ring shall be employed for positioning the course marker in response to a remote input signal. The servo shall be capable of receiving a change of command course at any rate between 0° to 60° per second without hunting. There shall be no appreciable hunt or oscillation in the servo system during steady state and dynamic operation and it shall position the course around the inside edge of the azimuth ring to within ±1° of the position of the test synchro. The design of the differential gearing shall result in operation of the course marker with respect to the azimuth ring, remote synchro command input, and knob command input as achieved on figure 3. The course marker servo shall conform to the electrical connector and pin wiring shown on figure 3.

3.16.2 Course marker signal input. The course marker servo system shall receive its input signal from a transmitting synchro device similar to the test transmitter described in 3.11. The course synchro device shall be a Clifton Precision transolver CDSH-10-AS-4, or equivalent, capable of operating as a CT and a transmitter as shown on figure 3. The XYZ load shall be equivalent to an EP AY500-3, or one exhibiting higher impedance characteristics.

3.16.3 Course datum control transformer. The course datum CT shall be Clifton Precision's CTC-8-A-4, or equivalent, and shall operate as shown on figure 3.

3.16.4 Course command and course datum control transformer zero. Both the course command and course datum CT's shall be zeroed as a standard test transmitter at an index reference of 270°. Both synchros can be zeroed in the same manner as the heading command CT and heading datum CT specified in 3.15.4 through 3.15.4e with course marker substituted for heading marker and with all references made to the course servo system (the course servo system should not be excited with power).

3.16.5 Course marker servo zero. The course command control transformer shall be connected to the standard test transmitter as per the XYZ notation given on figure 3. The excitation voltage of the test transmitter C to H shall be in phase with the servo power excitation voltage, pin AA to pin DD. The following conditions shall be satisfied:

a. The course marker shall position to 0° north on the

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azimuth ring when the dial of the standard test transmitter is at 0°.

b. The course marker shall rotate in an increasing heading indication on the azimuth ring (cw) when the dial of the test transmitter synchro is rotated in an increasing heading indication (dial ccw).

c. When the dial of the test transmitter synchro is rotated in an increasing heading indication (dial ccw), the voltage C to H' of the course command CT shall be in phase with the voltage from C to H of the standard test transmitter.

d. If the transmitter winding of the course command CT is excited with 26V ac so that the voltage C to H is in phase with power excitation ground to phase and the course pointer is set on 0° (north), then the voltage X to Y to Z will be in phase with C to H and voltage X to Y null. If the course pointer is rotated cw to east, the voltage X to Z shall increase before decreasing.

3.16.5.1 Course resolver. A resolver of any size with electrical characteristics of a type EP AY221-5-B, or equal, shall be connected to the course gear train to function as shown on figure 3 and zeroed at a scale reference of 0° and an index reference of 300°.

3.16.6 Rotor windings.

3.16.6.1 Rotor winding number 1. Rotor winding number 1 (RW-1) is defined as the leading winding which mechanically leads rotor winding number 2 (RW-2) by 90° in the direction of higher dial numbers with respect to the azimuth card. At the index reference point 300° when properly zeroed, stator winding 1 (SW-1) and RW-1 are directly coupled and stator winding 2 (SW-2) and RW-2 are directly coupled.

3.16.6.2 Stator winding number 1. The common pin of the rotor windings is designated as pin C. The two leads of a stator winding are arbitrarily labeled D and E and this stator winding is called SW-1.

3.16.6.3 When the course marker is rotated 300° against the azimuth card, a rotor lead shall be chosen and labeled A. This rotor winding shall be excited (C being the common ground) with 10V 400 cps ac and the resolver rotor shaft adjusted (with scale locked at 300°) respectively. The resolver rotor shaft shall be locked to scale and the scale rotated 90° in a positive direction to 30° against the azimuth card. The voltage A to C of RW-1 shall be removed and applied to B to C of RW-2. The voltage as

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read across D to E of SW-1 should be maximum and in phase with the applied voltage B to C of RW-2. If the voltage is out of phase, the labeling A and B of the rotor leads shall be reversed and the above zeroing procedure repeated.

3.16.6.4 The scale reading shall be set to 300° against the azimuth card and leads B and C of RW-2 excited. Letters F and G shall be assigned to the stator winding SW-2 so that the voltage measured from F to G is in phase with voltage B to C of RW-2.

3.16.6.5 Fine zeroing procedure. Fine zeroing can be accomplished by exciting leads B to C of RW-2 and nulling the voltage D to E of SW-1 by rotating resolver rotor shaft when scale reading is 300° against the azimuth card.

3.16.6.6 Course selection resolver. The course selector resolver shall have proper characteristics or additional provisions as required to permit use with VOR systems requiring 30 cps Omni Range Zero (ORZ). Proper ORZ at 30 cps shall be obtained when the resolver is electrically zeroed with 400 cps excitation at 300°. The ORZ leads shall be designated as leads H and C. The leads designated as A to G shall be connected to connector PT02H-22-55P as specified on figure 3. The VOR phase shifting component shall be connected to pin A of connector PT02H-22-55P as shown on figure 3.

3.17 Course bar. An indication of the displacement of the aircraft to the right or left of the selected course shall be provided by means of a course bar and dots in accordance with figure 2.

3.17.1 Course bar mechanism. Direct current deflection signals shall cause the course bar to move across the face of the instrument. The total movement of the course bar shall be 11/16 ±1/32 inch in both directions measured from the center of the instrument. A movement of 5/8 inch from center shall require 150 ua (microamperes) nominal. A dc polarity of "+" and "-" indicated as pins of figure 3 shall cause a deflection on the right when course arrow is on top. The course bar sensitivity shall be 75 ua per dot and range of displacement shall be +2 dots. The course bar mechanism shall have a resistance of 1,000 ohms ±3 p pins of figure 3 shall cause a deflection on the right when course arrow is on top. The course bar sensitivity shall be 75 ua per dot and range of displacement shall be +2 dots. The course bar mechanism shall have a resistance of 1,000 ohms ±3 percent and a damping ratio of not less than 1. The frequency response of the course bar mechanism shall be such that amplitude ratio of output to input shall not decrease by more than 3 decibels at 0.2 cps. The test limits at standard conditions for

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the course bar mechanism shall be as follows:

±1 dot deflection	±075 ua	±06 ua
±2 dots deflection	±150 ua	±12 ua

3.18 To-from arrow. A TO-FROM arrow shall be incorporated as shown on figure 2. The TO-FROM arrow shall operate from a meter movement having a resistance of 250 ohms maximum and 150 ohms minimum. The sensing will be such that with no signal applied, the arrow will not be visible. A signal with polarity as shown on figure 3 shall cause the TO-FROM arrow to appear on the same side as the course arrow. A signal with opposite polarity shall cause the TO-FROM arrow to appear on the tail side of the course arrow. A signal not greater than 225 ua shall cause the TO-FROM arrow to reach full displacement.

3.19 Display movement. The central portion of the display, including the course arrow, course bar, the TO-FROM arrow, and course deviation dots, shall rotate as a unit turning with the azimuth ring as the airplane's heading changes, in addition to the individual motions listed.

3.20 Meter movement bearings and pivots. Meter movements shall turn on hardened ground and highly polished pivot bearings, highly polished first quality sapphire bearings, or other materials that meet the performance requirements of this specification.

3.21 Balancing. The meter movements shall be balanced by an approved means (threaded nut or splice nuts, or a wire helix moving along a crown arm and counterweight arm) with easily accessible means for rebalancing. The use of solder, shellac, or similar means of holding balance weights will not be acceptable.

3.22 Bearing marker. A bearing marker servo system shall be incorporated for positioning the bearing pointer shown on figure 2 from a remote source similar to a standard test transmitter synchro. The servo system shall be similar to the servo driving the azimuth ring. The servo shall receive a change in bearing between 0° to 60° per second without hunting. There shall be no appreciable hunt or oscillation in the servo system and it shall position the bearing pointer around the azimuth ring to within ±1° of the position of the test synchro. The servo system shall be connected as specified on figure 3. The bearing servo loop shall position both the bearing marker and the reciprocal bearing marker.

3.22.1 Bearing pointer signal input. The bearing pointer servo system shall receive its input signal from a transmitting synchro

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device similar to the test transmitter described in 3.11. The bearing pointer CT shall be a type EP AY-500-5, or equal. A size 8 synchro CT with the general impedance of the stator windings as a type EP AY500-5, or equal, may be utilized.

3.22.2 Bearing pointer control transformer zero. The bearing pointer CT shall be zeroed at a scale reference of 180° (under the lubber line at the bottom of the display) and at an index reference of 90°.

a. With the standard test transmitter connected as per the XYZ notation on figure 3 and excited with voltage from C to H in phase with the servo excitation pins AA to BB, the transmitter dial shall be positioned to zero.

b. The bearing pointer shall position under the lubber line at the bottom of the display. The reciprocal bearing pointer shall be 180° from the bearing pointer for all transmitter inputs.

c. Rotation of the card on the test transmitting synchro to an increasing heading indication shall cause the bearing pointer to rotate cw.

d. If the rotor of the test transmitter synchro is rotated to an increasing heading indication the voltage C to H' of the bearing, CT shall be in the same phase sense as the voltage C to H of the standard test transmitter.

3.23 Distance display. A digital-type distance display shall be provided at the upper left of the instrument as specified on figure 2. The range of the display shall be 000 miles to 1,999 miles. The hundreds, tens, and units numerals shall be individually driven by receiver synchros with a 1:1 ratio. The units display shall have 1-mile and 1/2-mile index lines. The numerals shall be centered each 36° of synchro shaft rotation. The receiver synchros shall be Clifton CRC-8-A-1, or equal. The thousand digit shall be driven by a shutter showing black when normally unexcited and 1 when excited with 27V dc.

3.23.1 Distance synchro electrical zero. With a standard test transmitter connected as per the XYZ common with C notation and C to H of both the receiver synchros and transmitter synchro excited, the three synchros shall be zeroed as follows:

a. With the test transmitter rotor position set on 0°, the 0 shall be in position on the distance dial of each receiver synchro.

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b. Rotation of the test transmitter synchro to 36° shall result in 1 being positioned on the distance dial.

c. Increasing numerical dial readings on the receiver synchros shall be indicated for each 36° of the transmitter synchro rotation.

3.23.2 Distance shutter mechanism. A black, masked shutter shall be provided for obscuring the distance display. The shutter mechanism shall be excited by 28V dc. Voltage excitation shall cause the shutter to disappear from view. The units digit shall be only partially masked by the shutter.

3.24 Power. The HSI shall utilize only power that may be supplied by the 115V 400 cps, single phase, a-c aircraft electrical system in accordance with MIL-STD-704. Each of the four servo systems shall be excited with 115V 400 cps on the specific connector and pins specified on figure 3. In addition, 5V a-c power shall be supplied to the proper connector and pins of figure 3 for operation of the integral lighting.

3.24.1 Electrical power variation. The indications shall operate properly and continuously when the a-c voltage supplied to the indicator connector is in a range between 103 to 127V and at any frequency between 320 and 480 cps or any combination thereof, in addition to transient variation specified in MIL-STD-704.

3.25 HSI connector. An electrical connector shall be supplied on the HSI as shown on figure 1. The connector shall be a Scintilla Division, Bendix Aviation Corporation, part number PT02H-22-55P connector, or equal, and shall mate with part number PT06A-22-55S, or equal. All electrical connectors shall be in accordance with figure 3.

3.25.1 Protection of electrical connector. The electrical connector shall be protected from water vapors by the application of a cap and gasket conforming to Scintilla part number 10-123094-221, or equal.

3.26 Integral lighting. The HSI shall be integrally lighted with two parallel lighting circuits (a white lighting circuit and a red lighting circuit) as shown on figure 3. The red lighting system design shall be in accordance with MIL-L-25467 and the white lighting system design shall be in accordance with MIL-L-27160. When specified by the procuring activity, room shall be provided for the addition of a resistor. The lighting shall meet the specified requirements and tests without the use of resistors in the circuit. Data for the exact intensity level of the

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integral lighting shall be provided by the procuring activity. The lighting shall be subject to approval by the procuring activity. The integral lighting shall be comparable to and shall balance in brightness with prime standard HSI. The prime standard shall be a HSI conforming to the requirements of this specification or a lighted mockup of the HSI. The prime standard shall have applied to its lighting circuits the specific voltage which the procuring activity has established as being necessary to assure equality of presentation (from a lighting standpoint) with adjacent but dissimilar instruments. The contractor shall submit to the procuring activity the test procedure for determining compliance with this requirement.

3.27 Weight. The weight of the HSI shall not exceed 9.5 pounds.

3.28 Screw threads. Unless otherwise specified, the threads of all machine screws 0.060 or larger in diameter shall conform to MIL-S-7742.

3.29 Lubrication. The HSI shall be lubricated in accordance with MIL-STD-838.

3.30 Finishes and protective coatings.

3.30.1 Aluminum alloy parts. Aluminum parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

a. Dials, small holes, and case inserts need not be anodized.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

d. Castings containing nonalumimum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.30.2 Protective coating. Cadmium plating must be avoided when satisfactory alternative processes can be used. Parts previous requiring cadmium plating shall be coated with ion vapor

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deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity.

3.31 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.32 Workmanship. The HSI, including all parts and accessories, shall be so fabricated and finished as to assure freedom from blemishes, defects, burrs, sharp edges, accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts.

3.32.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.32.2 Cleaning. The HSI shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign materials removed during and after final assembly.

3.32.3 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the HSI, they shall be held or limited accordingly.

3.32.4 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.33 Installation.

3.33.1 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the HSI. They shall be round-head, brass machine screws and shall have a durable dull-black finish. Length shall be sufficient to permit mounting on a panel 3/16-inch thick. Screw threads shall be size 10-32NC-2A.

3.33.2 Envelope. An envelope containing the mounting screws shall be furnished with each HSI. The following information shall be printed on the face of the envelope:

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IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of test. The inspection and testing of the HSI shall be classified as follows:

a. First article tests.

b. Acceptance tests.

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25° C).

4.3.2 Voltage input. Unless otherwise specified, the HSI shall be tested with 115V 400 cps (nominal) single-phase power connected to the specific pins of the indicator connector.

4.3.3 Proper connection. When the HSI is referred to as being properly connected, it shall be understood that it is connected to a test fixture that is providing a lighting circuit voltage variable from 5 to approximately 1V, synchro signals for remote inputs to the servo systems, variable current d-c signals with reversible polarity to the meter movements, 27V dc for shutter excitation, and a-c power as required for operation of the servo systems.

4.3.4 Optional testing. The hunting, jumping and lag test required in 4.6.3 may be performed at the contractor's option, prior to final sealing of the indicator.

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4.3.5 Visual observation. Unless otherwise specified, all visual observations of HSI performance shall be made at a distance of 2 feet.

4.4 First article testing.

4.4.1 First article test samples. The first article test samples shall consist of three HSI's. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.4.2 First article required. Prior to actual procurement, the product which this specification covers shall pass the first article tests specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same first article tests.

4.4.3 First article tests. The first article tests shall consist of all the tests specified under 4.6 Test methods.

4.5 Acceptance tests. Acceptance tests shall consist of :

a. Individual tests

b. Sampling plans and tests

4.5.1 Individual tests. Each HSI shall be subjected to the following tests as described under 4.6 TEST METHODS:

a. Examination of product

b. Synchro and receiver zero

c. Servo operation

d. Servo sensitivity

e. Scale error

f. Servo followup rate and damping

g. Course bar mechanism

h. Balance

i. Accuracy

j. Stop test

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k. TO-FROM arrow

l. Distance display

m. Shutter

n. Sealing

o. Lighting

p. Fogging

q. Early failure detection

r. Control knobs

4.5.2 Sampling plans and tests.

4.5.2.1 Sampling plan A. HSI's selected at random in accordance with the schedule specified shall be subjected to the following tests as describer under 4.6 TEST METHODS:

Number of Items	Quantity of Items to be Tested
First 10	1
Next 90	2
Each additional 100	1

a. Individual tests

b. Power consumption

c. Voltage variation

d. Magnetic property

e. Acceleration

f. Dielectric

g. Vibration error

h. Oscillation and variation

i. Servo null position shift

j. Low temperature operation

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k. High temperature operation

l. Course bar mechanism

m. Coating

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 HSI's shall be selected at random from the first 15 items on the contract or order and subjected to the following tests in accordance with the schedule specified and as described under 4.6 TEST METHODS:

a. Sampling plan A tests

b. Radio noise interference

c. High temperature exposure

d. High altitude-low temperature

e. Temperature shock

f. Humidity

g. Fungus

h. Salt spray

i. Sand and dust

j. Vibration failure

k. Acceleration

l. Life

m. Sealing thermal shock

n. Cover glass abrasion

NOTE: At the option of the contractor, the tests may be divided into three groups as follows, to be conducted on three sets of HSI's, if a need exists to expedite tests. Each HSI shall meet the requirements of all the individual tests after completing the group of specific tests to which submitted. At the option of the procuring activity, group II testing may be conducted on empty cases which have all external parts installed or attached, as on a completed HSI case.

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Group I	Group II	Group III
Radio noise interference	Humidity	Vibration failure
High temperature exposure	Fungus	Acceleration
Low temperature	Salt spray	Life
Temperature shock	Sand and dust	
	Sealing thermal shock	
	Cover glass abrasion	

4.6 Test methods.

4.6.1 Examination of product. Each HSI shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Synchro and resolver zero. Each CT receiver synchro and resolver shall be checked for zero by the method specified in section 3.

4.6.3 Servo operation. The HSI shall be properly connected. The rotor of the calibrated synchro shall be rotated at constant speeds of 30° and 60° per second. All four servos shall be tested. The indicated azimuth heading, bearing, and course indications shall not differ from the calibrated synchro indication by more than ±2° at 30° per second and ±5° at 60° per second. The tests shall be conducted in both cw and ccw directions.

4.6.3.1 Hunting and jumping. There shall be no noticeable hunting or jumping when the standard calibrated synchro shaft is not rotated. Under dynamic conditions, the peak to peak amplitude of hunting and jumping shall not exceed 1/4° for either of the rate tests specified in 4.6.3.

4.6.3.2 Servo sensitivity. The HSI shall be properly connected. The rotor of the calibrated synchro shall be rotated 1° and there shall be a perceptible movement in each of the pointers and dial as their respective CT's are excited.

4.6.3.3 Scale error. The HSI shall be properly connected. The standard calibrated synchro shall be set on zero and the heading indication read and recorded. The calibrated synchro connected to the azimuth card shall be turned successively to headings of 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°, 270°, 300°, 330°, and 360° and the readings under the lubber line recorded. Any one reading shall not differ from the corresponding synchro heading by more than ±1°. An identical procedure shall be followed for the heading and course marker, except that the reading shall be with respect to the azimuth card. An identical

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procedure shall also be followed for the bearing pointer, except that the test transmitter will be set at 180° (bearing pointer will position at the top lubber line) and readings taken relative to this zero.

4.6.4 Servo follow-up rate and damping. The HSI shall be properly connected. The calibrated synchro connected to the azimuth card shall be de-energized, its rotor rotated 90° and re-energized. The time required for the azimuth ring to rotate between 5° and 85° of its original displacement from the calibrated synchro shall not exceed 1.5 seconds, and the total time before the azimuth ring comes to rest shall not exceed 2 seconds. An identical test shall be performed on each of the other servo systems.

4.6.5 Course bar mechanism. The course bar mechanism shall operate with the polarity and sensitivity specified in 3.17.1.

4.6.5.1 Balance test. A static balance test with the normal 1g force of gravity acting successively along each of the mutually perpendicular axis of the case shall not cause the bar to deviate from its center position more than 1/2 bar width.

4.6.5.2 Accuracy test. An accuracy test with the ±d-c polarity applied to the proper terminals shall result in the following:

±1 dot deflection ± 75 ua ± 6 ua
±2 dots deflection ±150 ua ±12 ua

The tolerances for the tests specified in 4.6.20, 4.6.21, 4.6.25.1, 4.6.25.2 and 4.6.27 shall be as follows:

±1 dot deflection ± 75 ±10 ua
±2 dots deflection ±150 ±17 ua

4.6.5.3 Stop test. Twice normal current shall be applied in each direction and slowly reduced. The pointer shall not stick.

4.6.6 TO-FROM arrow. The sensitivity of the TO-FROM arrow shall be checked by applying current in each direction as specified in 3.18. Double voltage shall be applied in both directions and slowly reduced. There shall be no sticking.

4.6.7 Distance display. Standard test transmitter synchros shall be applied to each receiver synchro. A scale error test shall be conducted for each numeral of the units, tens and hundreds counter. Each numeral shall center on even 36° ±4° increments of the test transmitter as specified in 3.23.1. A voltage of 27V dc shall be applied to the thousand shutter and

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the thousand digit shall appear. Counter motion shall be smooth with no sticking.

4.6.8 Shutter. Proper polarity of 28V dc shall be applied to the shutter mechanism. The shutter shall disappear from view.

4.6.9 Leakage rate. The HSI shall be tested for leakage by means of a mass-spectrometer-type helium leak detector. The initial maximum detected leak rate at a pressure differential of 1 atmosphere shall not permit more than 10 percent loss of the total filling medium after 1,000 hours.

4.6.10 Lighting. Each lighting circuit shall be properly connected individually and tested to show compliance with 3.26.

4.6.11 Fogging. The HSI shall be properly connected with power applied and placed in a +71° ±2° C controlled ambient for a minimum period of 1 hour. Upon completion of this period and still at the specified temperature, an ice cube shall be rubbed on the HSI cover glass for a period of from 1 to 2 minutes. The glass shall be carefully wiped dry (do not use compressed air) and the HSI inspected for evidence of water or oil fog on the cover glass, wedge, or display. Evidence of fogging shall be cause for rejection.

4.6.12 Early failure detection. Following final assembly of the HSI and the completion of the preceding individual tests, the HSI shall be properly connected with power applied to each of the four servo systems and the lighting circuits. The four servos and three distance counters shall be operated at 10° per second for approximately 720° and then alternated in the opposite direction. The course bar and TO-FROM arrow shall be driven through their full range one time per minute. The heading and course knobs shall be rotated for 5 revolutions in each direction every hour with the remote transmitter synchro source excited. The test shall be conducted for a minimum of 10 hours. No failure shall occur during this test.

4.6.13 Power consumption. The power consumption of the HSI shall be measured with suitable meters. The a-c consumption shall not exceed 50 va.

4.6.14 Voltage variation. The HSI shall be properly connected and the a-c voltage reduced to 103V and 320 cps. The indicator shall then meet all the individual tests, except that the scale error accuracy requirements may be increased from 1° to 2° and the servo operation requirements may be decreased from 10 to 20 percent between the range of 320 to 380 cps. The a-c voltage shall then be increased to 127V and 480 cps and the HSI

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shall meet all the individual tests.

4.6.15 Magnetic property. This test shall be made first with no power applied to the HSI and then repeated with the HSI operating on rated power. The HSI shall be held in various positions magnetically east or west of and 12 inches from the center of a free magnet, approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted. The maximum deflection of the magnet shall not exceed 5°. An aircraft compass with the compensating magnets removed may be used as the free magnet for this test.

4.6.16 Acceleration tests. The HSI shall be properly connected and power applied to the servos and meter movements. The HSI shall be accelerated along each axis with a force of 5g. The servoed pointers and azimuth card shall not vary by more than 1° from the initial indication and the deviation bar shall not vary by more than one bar width from its initial position. The TO-FROM arrow with no power applied shall not appear.

4.6.17 Dielectric. A potential of 250V dc shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.6.18 Vibration error. When mounted as specified in procedure IV of MIL-E-5272, properly connected, and subjected to vibration with a total amplitude between 0.003 and 0.005 inch, the HSI shall meet the following:

a. The maximum total amplitude of pointer oscillation, azimuth card oscillation, and deviation bar oscillation shall not exceed 1/2° in pointer and azimuth card or 1/2 deviation bar width.

b. The maximum variation of the pointers, azimuth card, and deviation bar, during vibration from the original indication before vibration throughout the frequency range, shall be noted and shall not exceed 1/2° in pointer and azimuth card or 1/2 of the deviation bar width.

4.6.18.1 Oscillation and variation. Oscillation and variation tests may be conducted at any indication.

4.6.19 Servo null position shift. The four servos shall be properly connected to standard test transmitters set at the zero indication. The input voltage and frequency shall be varied within the following limits, and combinations:

Frequency	320 to 480 cps
AC Voltage	103 to 127V

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The zero indication of the servoed pointers and azimuth card shall not shift more than 1° from the normal null position.

4.6.20 Low-temperature operation. The HSI shall be properly connected, except that no power shall be applied, and shall be subjected to a temperature of -54° ±2° C for a period of 4 hours. At the end of this period and while still at the low temperature, power shall be applied. There shall be no sticking of moveable component parts, and pointers shall move throughout their range of travel without sticking. There shall be no failure as a result of this test. There shall be a 5-minute warmup period after the application of power before data is taken. The HSI shall meet the tests specified in 4.6.3 through 4.6.9, except that an additional 15 percent shall be permitted on all tolerances specified.

4.6.21 High-temperature operation. With power applied, the HSI shall be properly connected, placed in a chamber, and maintained at a temperature of +71° ±2° C for a period of 4 hours. There shall be no sticking of moveable component parts, and pointers shall move throughout their range of travel without sticking. There shall be no failure as a result of this test. The HSI shall meet the tests specified in 4.6.3 through 4.6.9, except that an additional 15 percent shall be permitted on all tolerances specified.

4.6.22 Course bar mechanism. The course bar mechanism shall meet the dynamic response characteristics specified in 3.17.1.

4.6.23 Coatings. The coated sample required in 3.26.1 shall be evaluated to determine transmittance and reflectance of light emitted from a 2854° K source. The values shall not exceed those specified in section 3.

4.6.24 Radio noise interference. Radio noise interference test shall be in accordance with MIL-STD-826.

4.6.25 Environmental. The HSI shall be subjected to the following tests, conducted in accordance with the specified procedures of MIL-E-5272 and as specified herein.

4.6.25.1 High-temperature exposure. The HSI shall be properly connected and subjected to a high-temperature test in accordance with procedure II for a period of 24 hours, except that no power shall be applied. At the end of this period and while still at the high temperature, the HSI shall meet the tests specified in 4.6.3 through 4.6.9, except an additional 15 percent tolerance shall be permitted on all tolerances specified.

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4.6.25.2 Low-temperature high altitude. The low-temperature test shall be conducted in accordance with procedure II, except that examination shall be performed only at the end of the 72 hour period. The HSI shall be properly connected, but no power shall be applied. At the end of this period, the pressure in the chamber shall be reduced to the equivalent of 80,000 ±500 feet and power applied. There shall be no binding nor sticking at any indication. After the HSI has returned to room temperature, it shall meet all individual tests. Before data is taken, there shall be a 5-minute warmup period after the application of power.

4.6.25.3 Temperature shock. The temperature shock test shall be conducted in accordance with procedure I. After the test, the HSI shall meet all individual tests.

4.6.25.4 Humidity. The humidity test shall be in accordance with procedure I. The HSI and amplifier shall then meet the tests specified in 4.6.4 through 4.6.9. There shall be no deterioration that would adversely affect subsequent operation, and fungus growth shall not be supported.

4.6.25.6 Salt spray. The salt spray test shall be conducted as specified in procedure I for a period of 50 hours. At the end of the 50 hour period, the HSI shall meet the tests specified in 4.6.4 through 4.6.9. No damage that would adversely affect subsequent operation shall result from this test, and the tolerance specified under the individual tests shall apply.

4.6.25.7 Sand and dust. The sand and dust test shall be as specified in procedure I. At the end of this test, the HSI shall be removed and allowed to cool at room temperature. The HSI shall then meet the tests specified in 4.6.4 through 4.6.9.

4.6.25.8 Vibration failure. The indicator shall be subjected to a vibration test in accordance with procedure V. After the vibration test, the HSI shall meet the tests specified in 4.6.3 through 4.6.9. No looseness in the mechanism nor damage to any part of the HSI shall result from this test.

4.6.26 Acceleration. The HSI, nonoperating, shall be mounted in its normal operating position and accelerated first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The HSI shall be subjected to an acceleration of 20g in each of the axes for a period of 1 minute. At the end of this acceleration, the HSI shall meet the individual tests. No damage to the HSI shall result from this test.

4.6.27 Life. The HSI shall be subjected to a life test for

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1,000 hours. The HSI shall be properly connected and power supplied to each of the 4 servos and 3 distance counters shall be operated at 10° per second for approximately 720° and then alternately in the opposite direction. The course bar and TO-FROM arrow shall be driven through their ranges five times per minute. The distance shutter shall be operated for 30 minutes for every 1-hour period. The heading and course knobs shall be rotated for 10 revolutions in each direction every 30 minutes with the remote transmitting synchro source unexcited. There shall be no failure during the test. After the test, the HSI shall meet the individual tests. Failures that occur must be evaluated to determine whether they are purely random or are systematic due to a design or component weakness. Suitable corrective action must then be determined and action taken accordingly.

4.6.28 Sealing thermal shock. The HSI shall be immersed alternately in tap water maintained at 85° ±5° C and 5° ±4° C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes and not more than 5 seconds shall elapse between bath immersions. No damage to the hermetic seal shall result due to this test. Following the immersions, the HSI shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The initial maximum detector leak rate at a pressure differential of one atmosphere shall not permit more than 10 percent loss of the total filling medium after 1,000 hours.

4.6.29 Abrasion. An abrasive pad shall be pressed firmly against and pulled across the cover glass in two different directions in the form of a cross. The glass shall be scratched once in each direction. Pressure shall be applied gradually within the case until a 20-psi differential exists. This pressure shall be maintained for 30 minutes. There shall be no damage to the cover glass as a result of this test. The abrasive pad shall be a 600-grit sandpaper, 3/4 inch wide by 3 inches long, mounted on a wooden base.

4.6.30 Control knobs. The torque required to rotate the course and heading set knobs shall be measured throughout one revolution of the course arrow and the heading marker in both clockwise and counterclockwise directions and shall be within 2 to 10 inch ounces.

4.7 Rejection and retest. When one HSI selected from a production run fails to meet the specification, no HSIs still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

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4.7.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of HSIs on hand or later produced shall not be made until it is determined that the HSIs meet all the requirements of the specification.

4.7.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in HSIs already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.8 Packaging inspection.

4.8.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specifications and the marking requirements of MIL-STD-129.

4.8.2 First article. When the unit container is capable of serving as the shipping container, first article inspection and rough handling tests, as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A, B, or C as specified by the procuring activity.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit

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packs of one each in accordance with Method IC-1 of MIL-P-116. Cushioning material between bag and outer container shall be of a type, density, and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.4.2 of this specification.

5.1.2 Level B. Each unit shall be packaged as indicated for Level A except that a specification PPP-B-636 class domestic box shall be used as the outer container.

5.1.3 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B, or C as specified by the procuring activity.

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, specification PPP-B-601, plywood box, shall be used for the shipping container. Insofar as practical, exterior containers shall be of a uniform shape and size and of minimum cube and tare consistent with the protection required.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather-resistant.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The AQU-2/A HSI covered by this specification is intended for use in high performance aircraft with an integrated panel. The display will be controlled by reference

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and remote command sources providing transmitter synchro output signals and dc master movement signals.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Whether or not sampling plan B is to be omitted

c. Selection of applicable levels of packaging and packing and reinspection date markings.

6.3 First article. A First Article is required and shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Aquistion Regulation. First article tests are described in 4.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Definitions.

6.4.1 Index reference position. The index reference position of a synchro is that position specified in terms of dial reading where the voltage XY is zero and the stator voltage XZ and YZ is 180° out of phase with the rotor voltage HC. When the synchro is in this position, the voltage between H and X will be the sum of the rotor and stator voltages. The index reference position of the standard test transmitter is 0°, of the distance counter receiver synchros 0°, of the CIs in course and heading 270°, and the bearing pointer CT 90°.

6.4.2 Hermetic seal. A hermetic seal is a perfectly closed and airtight seal made between vitric, metallic, elastomeric and resinous materials. To meet the specified leak rate requirements, combinations of these materials may be used in areas approved by the procuring activity. A hermetic seal is not intended to include seals accomplished by gaskets.

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Air Force - 99

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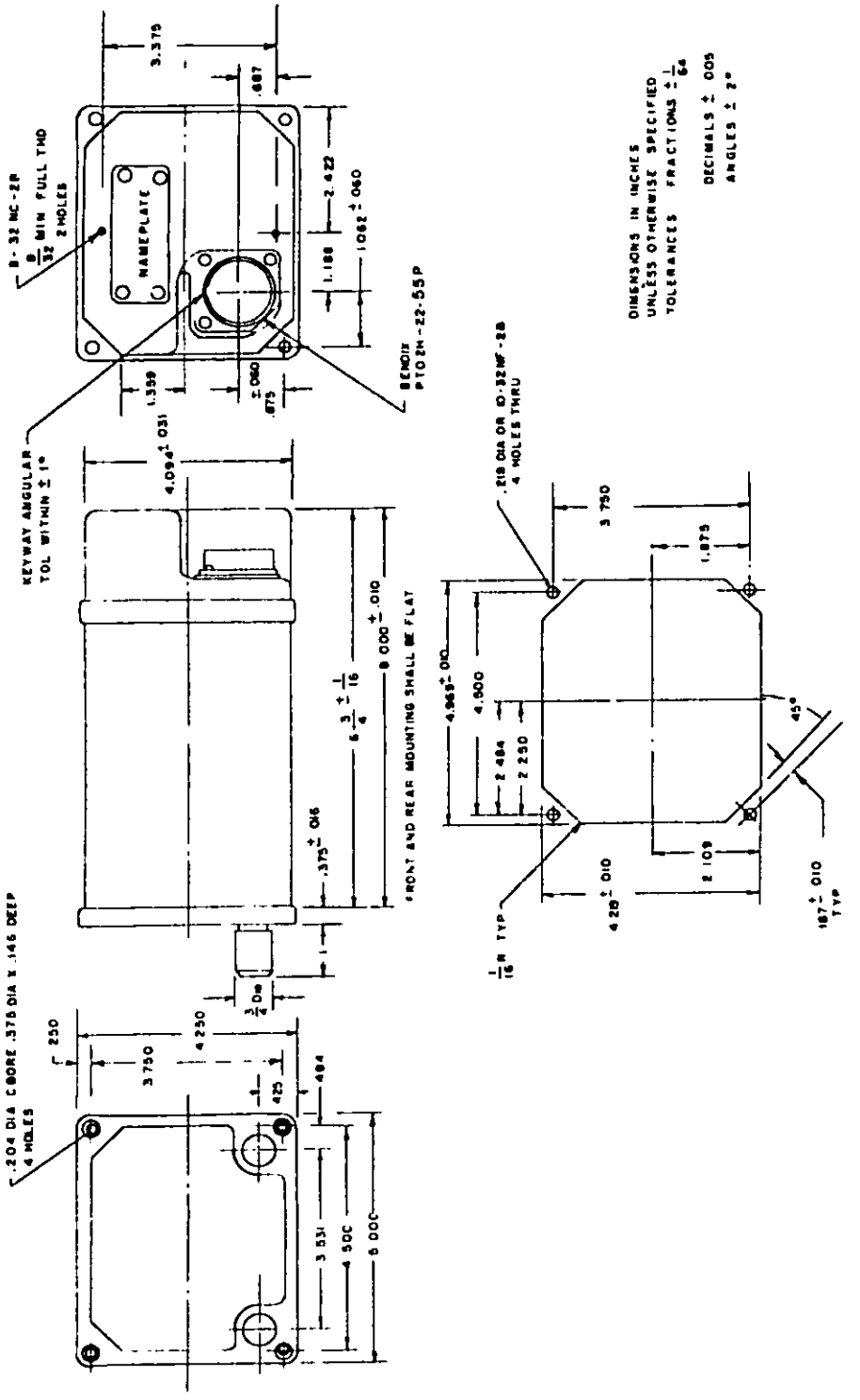
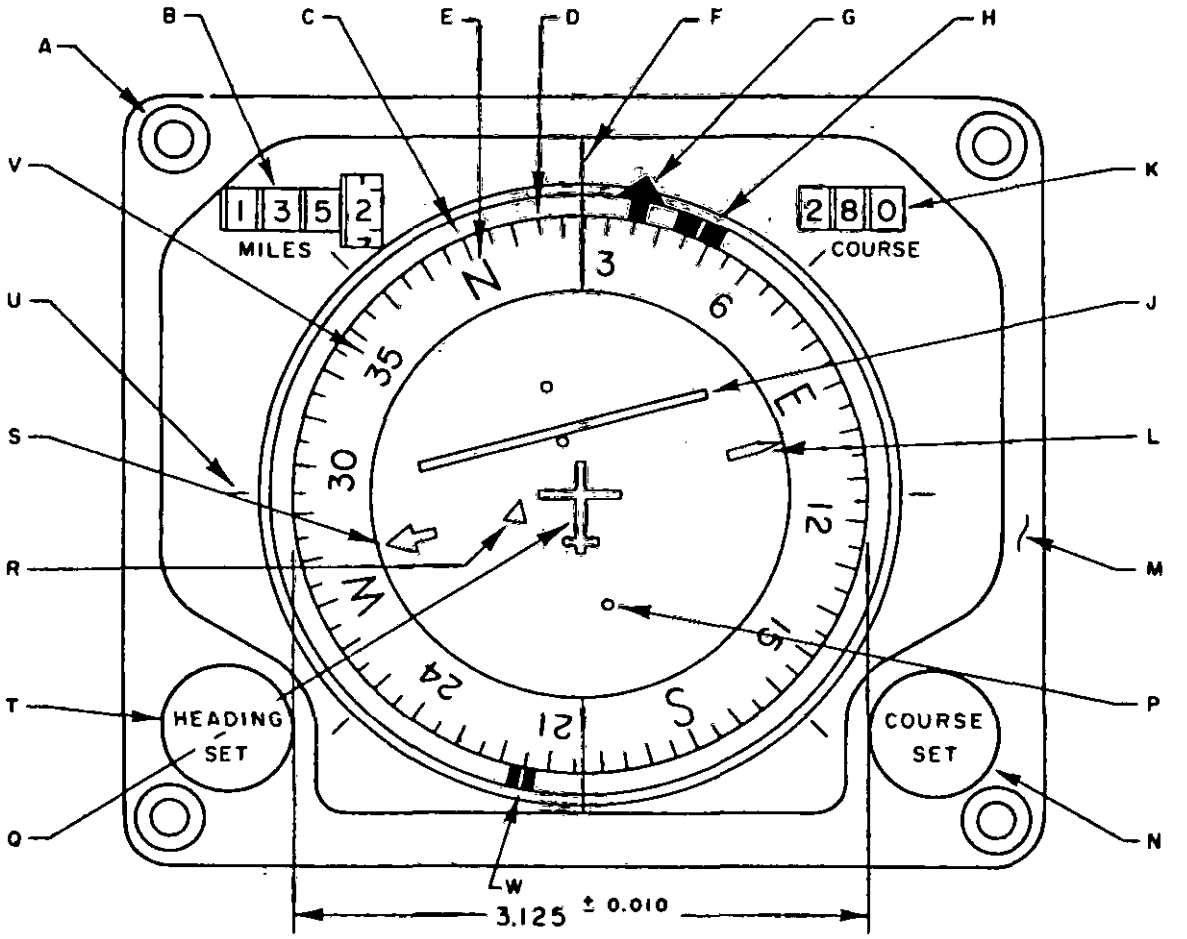


FIGURE 1. Case Dimensions

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A	MOUNTING HOLE	L	RECIPROCAL COURSE INDEX
B	DIGITAL DISTANCE DISPLAY	M	CASE FLANGE
C	10° INCREMENT MARKS	N	COURSE SET KNOB
D	5° INCREMENT MARKS	P	COURSE DEVIATION DOTS
E	CARDINAL DIRECTION	Q	AIRPLANE SYMBOL
F	LUBBER LINE	R	TO – FROM ARROW
G	BEARING POINTER	S	COURSE ARROW
H	HEADING MARKER	T	HEADING SET KNOB
J	COURSE DEVIATION BAR	U	FIXED MARKER
K	DIGITAL COURSE DISPLAY	V	NUMERALS
		W	RECIPROCAL BEARING POINTER

FIGURE 2. Dial

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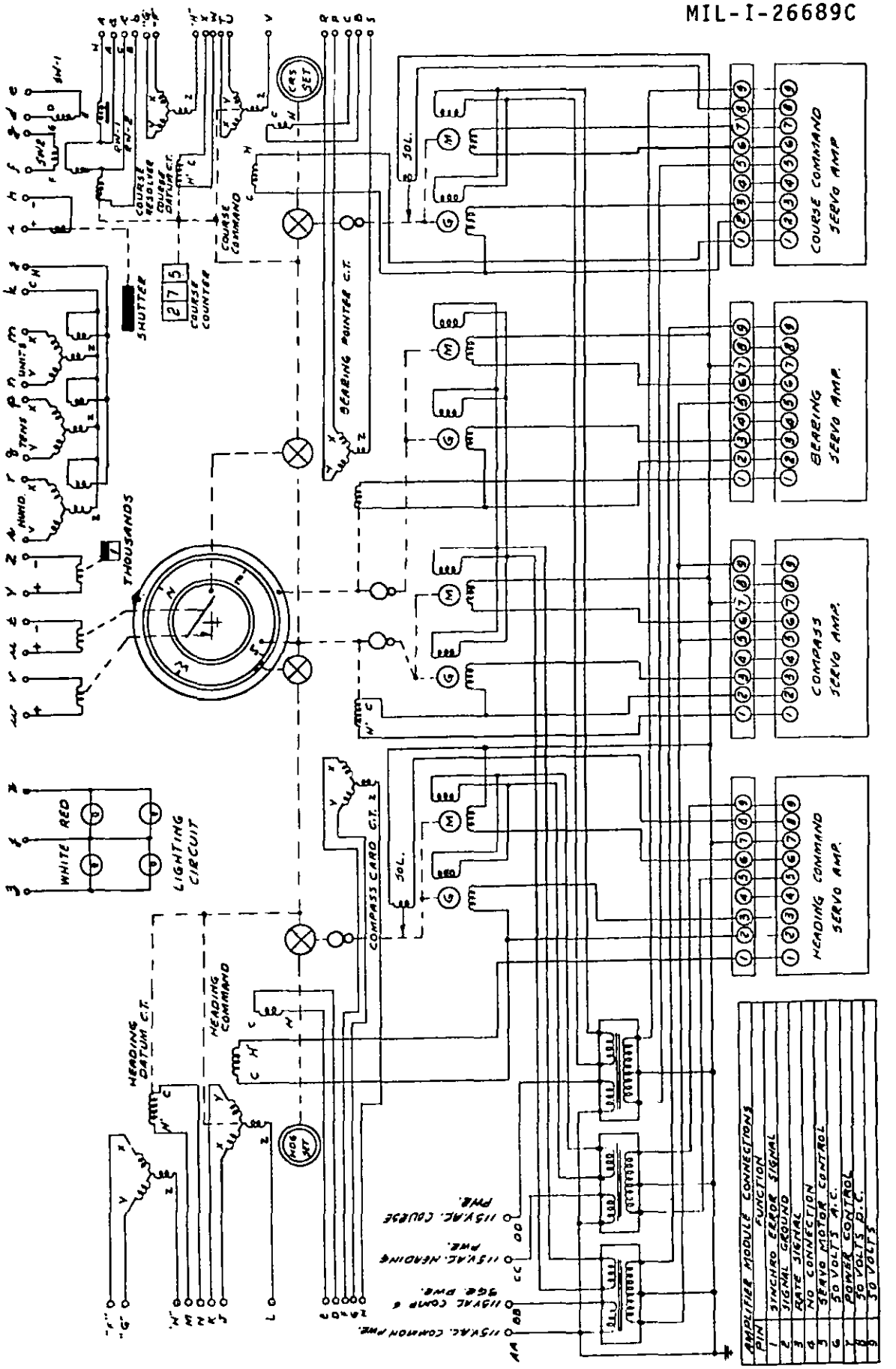
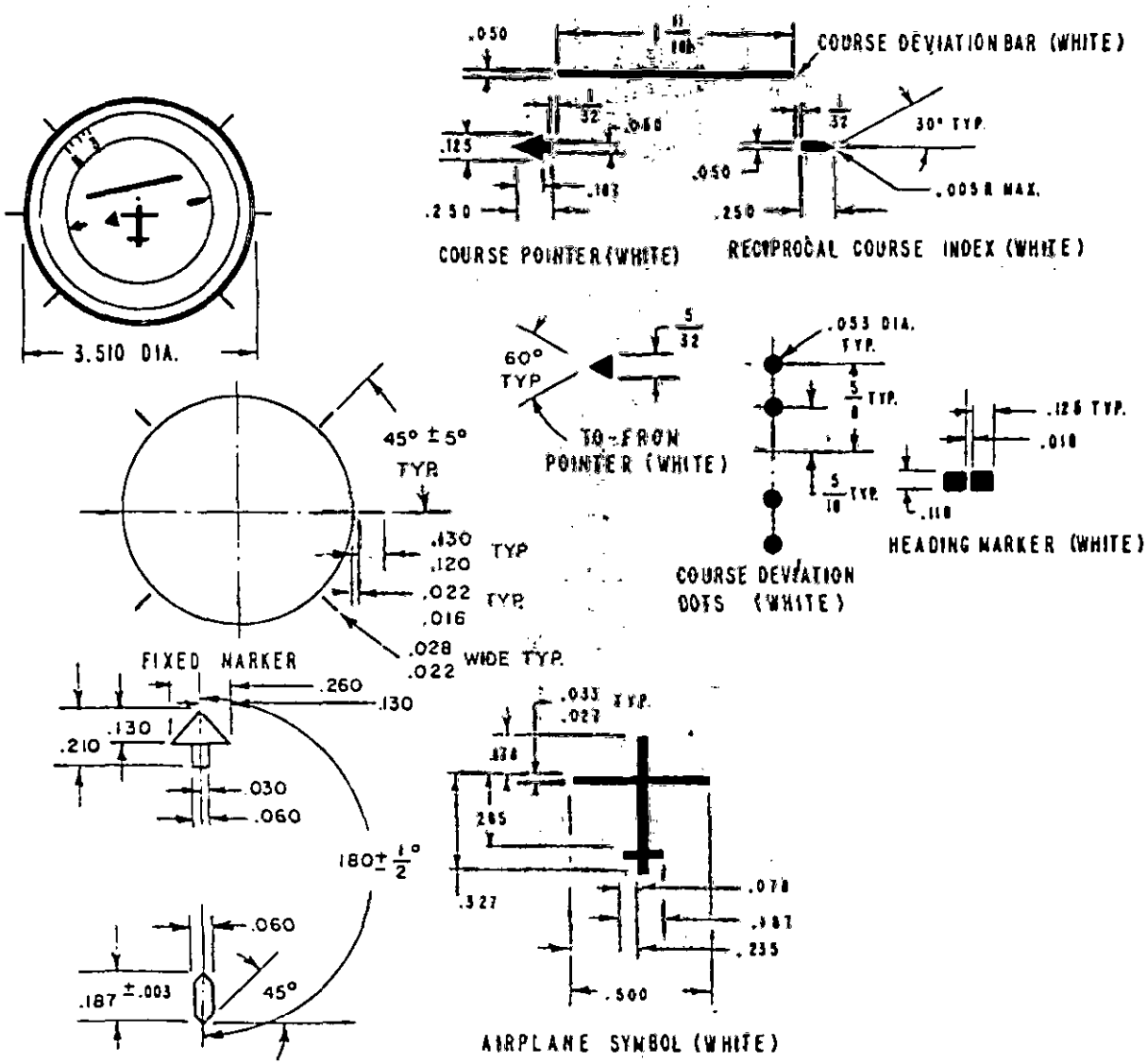


Figure 3. Wiring

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DIAL MARKING DATA

LETTERS: 1.155 R TO BOTTOM EDGE; .250 HIGH OVERALL; .032 - .040 LINE WIDTH.
NUMERALS: 1.217 R TO BOTTOM EDGE; .188 HIGH OVERALL; .024 - .030 LINE WIDTH.
10° LINES: 1.433 R TO INNER END AND EXTENDING WITHIN .005 OF O.D. OF PART; .016 - .020 LINE WIDTH, LENGTH .125 (REF).
5° LINES: 1.495 R TO INNER END EXTENDING TO WITHIN .005 OF O.D. OF PART; .008 - .010 LINE WIDTH, LENGTH .062 (REF).
NUMERALS AND LETTERS SHALL CONFORM TO MS33558.
DIMENSIONS IN INCHES: TOLERANCES:
UNLESS OTHERWISE SPECIFIED
DECIMALS: ± .005, FRACTIONS: ± 1/64, ANGLES ± 1°.

FIGURE 4. Display Component Dimensions

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Table I. Markings

	DIMENSIONS IN INCHES	COLOR	NOTES
A	0.204 Dia Counter Bore 0.375 Dia 0.146 Deep		
B	3/16 Numerals	White	
C	1/8 Long	White	
D	1/16 Long	White	
E	1/4 High X 0.030	White	
F	7/8 Long X 0.010	White	Positioned in same plane as Q.
G	45 Right Triangle	White	The base of the arrowhead shall be equivalent to the total width of the heading marker.
H	2-1/8 Squares	White	Spacing between squares shall be 0.018 +.003 -.000
J	1-3/4 X 0.050	White	Slot opening shall be held to a minimum.
K	3/16 Numerals	White	
L	1/4 Long X 0.050	White	
M	Per Figure	Black	
N	3/4 Dia X 1	Gray	
P	3/64 Dia - 5/16 on center	White	
Q	1/2 Airplane - Line Wt 0.030	White	
R	5/32 Equilateral Triangle	White	Opening shall be held to a minimum.
S	1/8 X 1/4	White	Positioned in same plane as course bar.
T	3/4 Dia X 1	Gray	
U	1/8 X 0.025	White	
V	0.20 High	White	

NOTE: 1. Stroke of numerals and graduations shall be 1/6 to 1/8 of height.
2. Width of numerals and letters shall be 3/5 of height.
3. Numerals and letters shall conform to MS 33558.

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